

Three Extreme Points Do Not Force an Isometry

A full counterexample to the open question in arXiv:2511.20345

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August 11, 2026

Status: candidate new full counterexample. An explicit rational three-dimensional polyhedral norm and an invertible rational operator give a negative answer. Every orthogonality-preservation claim reduces to entrywise nonnegativity of one of two displayed 3×3 matrices. Independent expert review is nevertheless recommended.

1 Source question

Manna, Mandal, Paul, and Sain ask on PDF page 16 of [1] whether, in an n -dimensional polyhedral Banach space, preservation of Birkhoff–James orthogonality at any n linearly independent extreme points forces a bounded linear operator to be a scalar multiple of an isometry.

(4.) Let $x_1 = (1, 0, 0)$, $x_2 = (1, \frac{1}{2}, 0)$, $x_3 = (0, 0, 1)$. Then $x_1, x_2, x_3 \in \partial \mathbb{X}$ are linearly independent eigenvectors of S . Clearly, x_1 is smooth and a level vector of S , but $x_1 \perp_B x_3$. Therefore, x_1, x_2, x_3 satisfy only conditions (i),(ii) and (iii).

We observed that for a finite-dimensional Banach space $\mathbb{X}$, if a linear operator preserves Birkhoff–James orthogonality on $\text{Ext } B_{\mathbb{X}}$ then the operator must be an isometry multiplied by a constant. The question that we raise here is that whether $\text{Ext } B_{\mathbb{X}}$ can be replaced by a proper subset of $\text{Ext } B_{\mathbb{X}}$. A more specific question is the following.

Open Question: *In an n -dimensional polyhedral Banach space, does the preservation of Birkhoff–James orthogonality by a bounded linear operator at any n linearly independent extreme points necessarily imply that the operator is a scalar multiple of an isometry?*

In case of an arbitrary Banach space the result is not necessarily correct, which follows from [12, Th. 2.33].

Figure 1: The exact open question in [1, p. 16].

Recall that $x \perp_B h$ means

$$\|x + \lambda h\| \geq \|x\| \quad (\lambda \in \mathbb{R}).$$

An operator T preserves Birkhoff–James orthogonality at x when $x \perp_B h$ implies $Tx \perp_B Th$ for every h .

2 Counterexample

Equip $X = \mathbb{R}^3$ with the norm

$$\|(a, b, c)\| = \max \left\{ |a|, |b|, |c|, \frac{1}{7} |-4a + 2b + 6c| \right\}. \quad (1)$$

This is a polyhedral Banach space: its unit ball is the cube $[-1, 1]^3$ intersected with the symmetric pair of slabs $|-4a + 2b + 6c| \leq 7$.

Consider

$$\begin{aligned} x_1 &= \left(1, -1, -\frac{1}{6}\right), & x_2 &= \left(\frac{3}{4}, 1, -1\right), & x_3 &= \left(-1, -1, \frac{5}{6}\right), \\ y_1 &= \left(1, 1, -\frac{5}{6}\right), & y_2 &= (1, -1, 1), & y_3 &= \left(-1, 1, \frac{1}{6}\right), \end{aligned} \quad (2)$$

and the invertible operator

$$T = \frac{1}{7} \begin{pmatrix} 4 & -2 & -6 \\ 0 & -7 & 0 \\ -16 & -6 & -25 \end{pmatrix}. \quad (3)$$

Theorem 1 (Negative answer to the source question). *The points x_1, x_2, x_3 are linearly independent extreme points of B_X . The operator T preserves Birkhoff–James orthogonality at each of them, but T is not a scalar multiple of an isometry. Consequently the open question in [1, p. 16] has a negative answer, already in dimension three.*

The rest of the packet verifies every assertion in the theorem.

3 An active-normal criterion

Write

$$e_1 = (1, 0, 0), \quad e_2 = (0, 1, 0), \quad e_3 = (0, 0, 1), \quad f = \frac{1}{7}(-4, 2, 6),$$

viewing these row vectors as functionals. The norm (1) is $\max\{|e_1(\cdot)|, |e_2(\cdot)|, |e_3(\cdot)|, |f(\cdot)|\}$.

Lemma 1 (Strict-sign cone test). *Let a simple vertex u of the unit ball of a three-dimensional polyhedral norm have active norm-one supporting functionals g_1, g_2, g_3 , and put*

$$G_u = \begin{pmatrix} g_1 \\ g_2 \\ g_3 \end{pmatrix}.$$

Then $u \not\perp_B h$ if and only if the three entries of $G_u h$ have the same strict sign.

Suppose S is invertible, $Su = v$ is another simple unit vertex, and

$$A = G_u S^{-1} G_v^{-1} \quad (4)$$

is entrywise nonnegative with no zero row. Then S preserves Birkhoff–James orthogonality at u .

Proof. The supporting-functional characterization of Birkhoff–James orthogonality says

$$u \perp_B h \iff \varphi(h) = 0 \text{ for some } \varphi \in J(u).$$

At a simple vertex, $J(u) = \text{conv}\{g_1, g_2, g_3\}$. Hence such a convex combination vanishes on h precisely when

$$\min_k g_k(h) \leq 0 \leq \max_k g_k(h).$$

Its negation is the strict-sign assertion.

For the second claim, argue contrapositively. If $v \not\prec_B Sh$, then $z = G_v Sh$ is either strictly positive or strictly negative entrywise. Since

$$G_u h = G_u S^{-1} G_v^{-1} z = Az,$$

entrywise nonnegativity and the absence of zero rows imply that Az has the same strict sign. Thus $u \not\prec_B h$. $\square$

4 Exact verification

The signed active-functional matrices at the six points in (2) are

$$\begin{aligned} G_{x_1} &= \begin{pmatrix} e_1 \\ -e_2 \\ -f \end{pmatrix}, & G_{x_2} &= \begin{pmatrix} e_2 \\ -e_3 \\ -f \end{pmatrix}, & G_{x_3} &= \begin{pmatrix} -e_1 \\ -e_2 \\ f \end{pmatrix}, \\ G_{y_1} &= \begin{pmatrix} e_1 \\ e_2 \\ -f \end{pmatrix}, & G_{y_2} &= \begin{pmatrix} e_1 \\ -e_2 \\ e_3 \end{pmatrix}, & G_{y_3} &= \begin{pmatrix} -e_1 \\ e_2 \\ f \end{pmatrix}. \end{aligned} \tag{5}$$

For example, $e_1(x_1) = (-e_2)(x_1) = (-f)(x_1) = 1$, while $|e_3(x_1)| = 1/6 < 1$. The other five checks are identical. Each displayed triple of active functionals is linearly independent, so all six points are simple extreme points of B_X .

Let X_0 and Y_0 be the matrices whose columns are the x_i and y_i , respectively. Direct arithmetic gives

$$\det X_0 = -\frac{7}{12}, \quad TX_0 = Y_0, \quad \det T = 4. \tag{6}$$

Thus the x_i are linearly independent, $Tx_i = y_i$, and T is invertible. For reproducibility,

$$T^{-1} = \begin{pmatrix} \frac{25}{28} & -\frac{1}{14} & -\frac{3}{14} \\ 0 & -1 & 0 \\ -\frac{4}{7} & \frac{2}{7} & -\frac{1}{7} \end{pmatrix}. \tag{7}$$

Substitution of (5) and (7) yields

$$G_{x_1} T^{-1} G_{y_1}^{-1} = G_{x_3} T^{-1} G_{y_3}^{-1} = \begin{pmatrix} \frac{3}{4} & 0 & \frac{1}{4} \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \tag{8}$$

and

$$G_{x_2} T^{-1} G_{y_2}^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ \frac{4}{7} & \frac{2}{7} & \frac{1}{7} \\ 1 & 0 & 0 \end{pmatrix}. \tag{9}$$

Both matrices are entrywise nonnegative and every row is nonzero. Lemma 1, applied three times, proves that T preserves Birkhoff–James orthogonality at x_1, x_2, x_3 .

It remains only to rule out a scalar isometry. The point $p = (1, 1, 1)$ is another unit vertex because $|f(p)| = 4/7$. But

$$Tp = \left(-\frac{4}{7}, -1, -\frac{47}{7}\right), \quad f(Tp) = -\frac{40}{7}, \quad \|Tp\| = \frac{47}{7}. \quad (10)$$

On the other hand, $\|x_i\| = \|Tx_i\| = \|y_i\| = 1$. If $T = cU$ for a linear isometry U , any one of the x_i would force $|c| = 1$, contradicting (10). This completes the proof of Theorem 1.

5 Independent checks and scope

1. **Full local preservation.** The proof uses the complete set $x_i^{\perp B}$ through Lemma 1; it is not merely preservation with respect to one supporting hyperplane. This distinction is why the obvious diagonal examples in ℓ_1^3 do not answer the source question.
2. **No numerical tolerance.** Every coordinate, norm, determinant, and cone matrix in the final counterexample is rational. The exploratory search was numerical, but the proof is entirely exact.
3. **Minimal claimed scope.** This disproves the universal statement in dimension three. It does not address whether special classes such as regular polygons, ℓ_1^n , or ℓ_∞^n admit positive theorems; the literature cited by the source contains such rigidity results.
4. **Reproducibility.** The accompanying attempt folder contains the polygon, cube, and general polyhedron cone-search scripts. Their only role was discovery and stress testing; no computation is needed to validate equations (8)–(10).

6 Novelty search and review recommendation

A bounded search on August 11, 2026 used the exact question, arXiv identifier, and combinations of “linearly independent extreme points”, “polyhedral Banach space”, and “Birkhoff–James orthogonality preservation”. It found the source, the authors’ earlier refinement paper, and positive results for all extreme points and several classical polyhedral spaces, but no later resolution of this exact question. The counterexample is therefore labeled a candidate new full negative answer.

Human review should verify the strict-sign formulation of Lemma 1, the six active-functional lists in (5), and the novelty search. The remaining verification is elementary matrix arithmetic.

References

- [1] J. Manna, K. Mandal, K. Paul, and D. Sain, *An improvement of the Blanco–Koldobsky–Turnšek characterization of isometries*, arXiv:2511.20345 (2025).